Student ID (*Matricola*)

0	0	0	0	0	0
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

Computing Infrastructures

Course 095897

M. Gribaudo, R.Mirandola

07-07-2017

Last Name / Cognome:

. . . .

First Name / Nome:

. . . .

Answers must be given exclusively on the answer sheet (last sheet): DO NOT FILL ANY BOX IN THIS SHEET

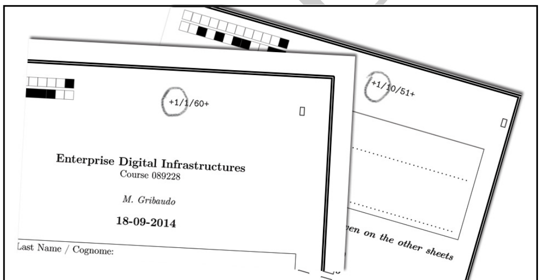
Students must use pen (black or blue) to mark answers (no pencil).
Students are permitted to use a non-programmable calculator.

Students are NOT permitted to copy anyone else's answers, pass notes amongst themselves, or engage in other forms of misconduct at any time during the exam.

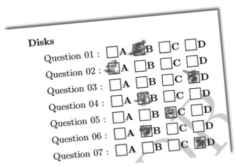
Students are NOT permitted to use mobile phones and similar connected devices.

Scores for the multiple-choice part: correct answers +1 point, unanswered questions 0 points, wrong answers -0.333 points.

You cannot keep a copy of the exam when you leave the room.

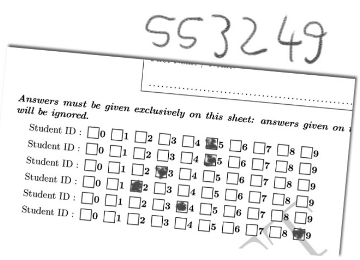


Check that the first number in the text and in the answer sheet is the same



Do not use crosses to mark the answers!

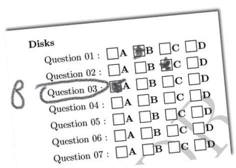
Completely fill the box!



Enter the numbers of your "Student ID" from left to right

If you make a mistake:

1) circle the word "Question"
2) write the correct answer to its side



Big Data - Code - (for all students *EXCEPT* geomatic students, coming from the BSC in Environment Engineering) - (4 points)

The following Apache Spark code processes the data coming from Stack Overflow.

```
import collections



Question = collections.namedtuple("Question", """id id_user title text keywords
                                              views votes""")
Answer = collections.namedtuple("Answer", "id id_question id_user text")
User = collections.namedtuple("User", "id reputation profile")

q1= Question(1,1,"Cassandra Upsert not working on conditional writes",
             """I made a conditional insert (if not exists) \\\
             statement using DataStax java driver but it doesn't work""",
             "Java Cassandra DataStax", 1, 0)
q2= Question(2,1,"New Spark 2.2 Cassandra Connector",
             """ Tried to run the new connector to Spark 2.2 got
             error code 99129 who can be of help?""",
             "Spark Cassandra", 2, 3)
u1= User(1, 1, "I'm an independent programmer, 8 years expertise in Java dev");
u2= User(2, 5, "I'm Matei, Spark creator");
u3= User(3, 5, "I'm Guido, Python benevolent dictator");

a1= Answer(1,1,2,"""I think there is still a problem in DataStax connector,
                   try to use the one at this link XXX""")
a2= Answer(2,2,2,"Did you check server IP and Scala version?")
a3= Answer(3,2,3,"""I think you are using Python 2.7, while the new
                  API is for Python 3.0""")

questionsRDD=sc.parallelize([q1,q2])
usersRDD=sc.parallelize([u1,u2,u3])
answersRDD=sc.parallelize([a1,a2,a3])


def get_keywords(s):
    return s.split(" ")


#Select the users with the largest reputation providing also their profile
max_reputation=usersRDD.map(lambda u: u.reputation).max()
print max_reputation

# FILL IN
power_usersRDD.collect()

#Select the keywords of the questions answered by the power users

#Select power users' answers first
power_users_ID=power_usersRDD.map(lambda u: u[0]).collect()
power_answersRDD=answersRDD.filter(lambda a: a.id_user in power_users_ID)
power_answersRDD.collect()
```

```
51 power_answersRDD_questions_IDs=power_answersRDD.
52   map(lambda a: a.id_question).distinct(). collect ()
53 print power_answersRDD_questions_IDs
54
55 print questionsRDD.filter(lambda q: q.id in power_answersRDD_questions_IDs).
56             #FILL IN
57
58 #provide questions and answers text
59 #FILL IN
```

Question 1 Complete line 41.

- [A] power_usersRDD=usersRDD.filter(lambda u: u[2]==max_reputation).map(lambda u:(u[1],u[-1]))
- [B] power_usersRDD=usersRDD.filter(lambda u: u.reputation==max_reputation).map(lambda u:(u.id,u.profile))
- [C] power_usersRDD=usersRDD.filter(lambda u: u[1]==max_reputation).map(lambda u:(u[1],u[-1]))
- [D] power_usersRDD=usersRDD.filter(lambda u: u.reputation==max_reputation).flatMap(lambda u:(u.id,u.profile))

SOLUTION:
power_usersRDD=usersRDD.filter(lambda u: u.reputation==max_reputation).map(lambda u:(u.id,u.profile))

Question 2 Complete line 56.

- [A] flatMap(lambda q: get_keywords(q.keywords)).collect().distinct()
- [B] flatMap(lambda q: get_keywords(q.keywords)).distinct().collect()
- [C] reduce(lambda q: get_keywords(q.keywords)).collect()
- [D] map(lambda q: get_keywords(q.keywords)).distinct().collect()

SOLUTION:
flatMap(lambda q: get_keywords(q.keywords)).distinct().collect()

Question 3 Complete line 59?

- [A] questionsRDD.join(answersRDD).flatMap(lambda x: x[1]).collect()
- [B] questionsRDD.join(answersRDD).map(lambda x:(x[1][0],[1][1])).collect()
- [C] questionsRDD.map(lambda q: (q.id, q.text)).join(answersRDD. map(lambda a: (a.id_question, a.text))).map(lambda x: x[1]).collect()
- [D] questionsRDD.join(answersRDD).map(lambda x: x[1]).collect()

SOLUTION:
questionsRDD.map(lambda q: (q.id, q.text)).join(answersRDD. map(lambda a: (a.id_question, a.text))).map(lambda x: x[1]).collect()

Question 4 Which is the type of the object returned at line 38?.

- [A] string
- [B] RDD
- [C] int
- [D] char

SOLUTION: int

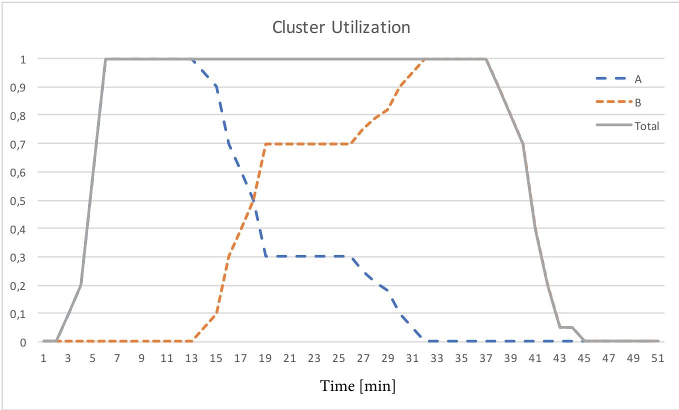


Figure 1: Cluster utilization under Capacity Scheduler.

Big Data - Theory - (for all students) - (4 points)

Describe the YARN *capacity* scheduler. Given the resource usage plot reported in the Figure, what is the configuration (in terms of pre-emption and work-conserving mode) for the two applications?

See BigDataIntro slide deck, slides 87 and 90.

The cluster is configured in work-conserving mode (each application uses the cluster at full capacity). Cluster capacity is split 30-70%, between application A and B, respectively. No pre-emption is adopted (application B takes a lot of time to reach its 70% capacity).

Big Data - Alt. - (in alternative to Exercise Big Data - Code, ***ONLY*** for geomatic students, coming from the BSC in Environment Engineering) - (4 points)

Describe Spark Streaming. Which are the main innovations introduced by the Structured Streaming API?

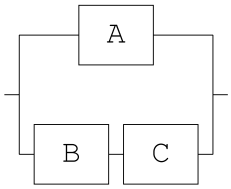
See Spark slide deck, slides 108-125.

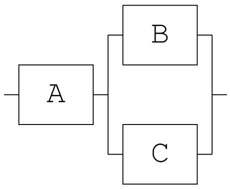
Dependability - (6 points)

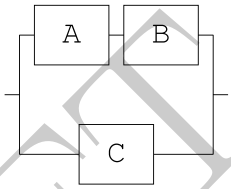
A simple storage system is composed by three components: a usb disk (A), a WiFi connection (B) and 4G cellular connection (C). The system is working when it can save a copy of a file on both the local (USB connected) disk, and in the cloud using a remote connection. All components are all characterized by the same $MTTR = 10$ days. The mean time to failure of the USB disk is $MTTF_A = 1000$ days. The mean time to failure of the WiFi connection is $MTTF_B = 50$ days. The mean time to failure of the 4G is $10\times$ times the one of the WiFi.

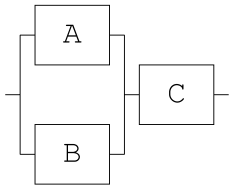
SOLUTION:
The availability of the components are:
$A_A = \dfrac{1000}{1000+10} = 0.9901$, $A_B = \dfrac{50}{50+10} = 0.83333$ and $A_C = \dfrac{500}{500+10} = 0.98039$

Question 5 An RDB (Reliability Block Diagram) describing the system is the following:

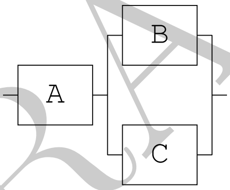
A 

B 

C 

D 

SOLUTION:



Question 6 The availability of the system is:

A $A = 0.98686$ B $A = 0.97877$ C $A = 0.99657$ D $A = 0.99819$

SOLUTION:
$A_A \cdot (1 - (1 - A_B) \cdot (1 - A_C)) = 0.98686$

Question 7 The $MTTR$ of the network component of the system (B and C) is:

A $MTTR_{BC} = 504.55$ days
C $MTTR_{BC} = 15$ days
B $MTTR_{BC} = 5$ days
D $MTTR_{BC} = 3.3333$ days

SOLUTION:
components have all the same $MTTR$, so $MTTR_{BC} = MTTR/2 = 5$ days.

Question 8 The $MTTF$ of the network component of the system (B and C) is

A $MTTF_{BC} = 15$ days
C $MTTF_{BC} = 504.55$ days
B $MTTF_{BC} = 1.6541$ days
D $MTTF_{BC} = 1525$ days

SOLUTION:
The availability of the parallel of components B and C is:
$A_{BC} = (1 - (1 - A_B) \cdot (1 - A_C)) = 0.99673$.
Using $MTTR_{BC} = 5$ days computed in the previous question and inverting the availability formula we have:
$MTTF_{BC} = \dfrac{A_{BC}}{1 - A_{BC}} MTTR_{BC} = 1525$ days.

Question 9 Now suppose that failures cannot be repaired. The reliability of the system at $T = 30$ days is:

A $R(30) = 0.92921$ B $R(30) = 0.97278$ C $R(30) = 0.94495$ D $R(30) = 0.98572$

SOLUTION:
$R(30) = R_A(30) \cdot (1-(1-R_B(30)) \cdot (1-R_C(30))) = e^{-\frac{30}{1000}} \cdot (1-(1-e^{-\frac{30}{50}}) \cdot (1-e^{-\frac{30}{500}})) = 0.94495$

Question 10 Consider again the repairable system. The user would like to add a third network connection (component D), in parallel with the other two, to reach a system availability of 0.99. The new component must be characterized by an availability of at least:

A $A_D = 0.9694$

B For this type of system, it is impossible to reach the target availability

C This system already reaches the target availability without requiring an extra component

D $A_D = 0.97877$

SOLUTION:

$$A_A \cdot (1-(1-A_B) \cdot (1-A_C) \cdot (1-A_D)) \geq 0.99$$
$$1-(1-A_B) \cdot (1-A_C) \cdot (1-A_D)) \geq \frac{0.99}{A_A}$$
$$(1-A_B) \cdot (1-A_C) \cdot (1-A_D)) \leq 1-\frac{0.99}{A_A}$$
$$1-A_D \leq \frac{1-\frac{0.99}{A_A}}{(1-A_B) \cdot (1-A_C)}$$
$$A_D \geq 1-\frac{1-\frac{0.99}{A_A}}{(1-A_B) \cdot (1-A_C)} = 0.9694$$

Cloud And Virtualization - (6 points)

Question 11 A program running at the Application Binary Interface (ABI) level:

- A Uses the user instructions of a CPU to do standard computation, and uses protected CPU instruction to access the hardware resources
- B Directly uses the user instructions of a CPU, but calls O.S. procedures to access the hardware resources
- C Uses emulation to run all instructions
- D Directly uses the user instructions of a CPU, but accesses the hardware resources through the Trap-and-Emulate technique

SOLUTION:
Directly use the user instructions of a CPU, but call O.S. procedures to access the hardware resources. See Slide 10 of L02 - Virtualization Part I

Question 12 Which of the following properties of virtualization prevents users of a VM to access the memory of other VMs running on the same physical host:

- A Isolation
- B Encapsulation
- C HW-independence
- D Partitioning

SOLUTION:
Isolation: see slide 57 of "L02 - Virtualization Part I"

Question 13 A end-user of a cloud application (i.e. a Costumer Relation Management application) typically:

- A Uses a SaaS developed with a PaaS and running on a IaaS
- B Uses a PaaS developed on a SaaS and running on a IaaS
- C Uses a IaaS developed on a SaaS and running with a PaaS
- D Uses a IaaS developed with a PaaS and running on a SaaS

SOLUTION:
Uses a SaaS developed with a PaaS and running on a IaaS, since Software as a Service (SaaS) allows the end-user to run the software. A SaaS application can be developed in a simpler way using a Platform as a Service (PaaS) framework. The machines used by the PaaS framework can be better provisioned using an Infrastructure as a Service (IaaS) model.

Question 14 A cloud in which a company holds its own cloud infrastructure, but can use virtual machines from other fellow private clouds with which some partnership relations exists is a:

- A Hybrid Cloud
- B Private Cloud
- C Public Cloud
- D Community Cloud

SOLUTION:
Community Cloud: see slides 34 of "L5 - Cloud Computing"

Network Virtualization

A physical server, whose IP address is 192.168.131.26, is running two VMs (called VM_1 and VM_2), both with a VNA configured in NAT mode. Both VMs are running a web server on Port 80.

Question 15 A typical set of addresses assigned to the two VMs is the:

- A VM_1: 10.0.2.15, VM_2: 10.0.2.15
- B VM_1: 192.168.131.27, VM_2: 192.168.131.28
- C VM_1: 192.168.131.27, VM_2: 10.0.2.15
- D VM_1: 192.168.131.27, VM_2: 192.168.131.27

SOLUTION:
VM_1: 10.0.2.15, VM_2: 10.0.2.15, since both for NAT VANs, all VMs are in different subnetworks.

Question 16 Which address could an application running on VM_2 use to access the web server:

- A http://10.0.2.15:8080, with a port-forwarding rule from port 8080 of the host to port 80 of VM_1
- B http://10.0.2.15:80
- C http://192.168.131.26:8080, with a port-forwarding rule from port 8080 of the host to port 80 of VM_1
- D http://192.168.131.27:80

SOLUTION:
http://192.168.131.26:8080, with a port-forwarding rule (PFR) from port 8080 of the host to port 80 of VM_1, since in NAT mode PFRs allows to open ports on the host and forward packet to the corresponding VM.

Disks - (4 points)

Consider the following set of six disks connected in RAID 6, with generator $g = 3$, $MTTF = 1515$ days, $MTTR = 15$ days, rotating at 12000 RPM, with an average seek of 2 milliseconds, transfer time of 256 MB / second:

Block A1	Block A2	Block A3	Block A4	Block A5	Block A6
D0: 1	D1:3	D2:	D3:	P: 11	Q: 163

Figure 2: A RAID 6 configuration.

Question 17 What data is contained in blocks $A3$ and $A4$?

- [A] Cannot be computed
- [B] $D_2 = 2$, $D_3 = 5$
- [C] $D_2 = 6$, $D_3 = 3$
- [D] $D_2 = 2$, $Q = 5$

SOLUTION:
$1 + 3 + D_2 + D_3 = 11$ and $1 + 3 \cdot 3 + 9 \cdot D_2 + 27 \cdot D_3 = 163$.
$D_2 = 7 - D_3$ and $9(7 - D_3) + 27 \cdot D_3 = 153$
$D_3 = 90/18 = 5$, $D_2 = 7 - 5 = 2$.

Question 18 The average time required to read a 4 KB block on one of the disk of the array independently:

- [A] 3.215 ms
- [B] 4.515 ms
- [C] 20.125 ms
- [D] 9.03 ms

SOLUTION:
The total average transfer time is:
$T_a = T_l + T_s + F/r_t = \frac{60000}{2 \cdot 12000} + 2 + 4/(256 * 1024) * 1000 = 4.515$ ms

Question 19 The average time before experiencing data loss is approximately

- [A] 815 years
- [B] 5 years
- [C] 705 years
- [D] 507 years

SOLUTION:
MTTDL is computed as:
$\frac{2 \cdot MTTF^3}{G \cdot (G-1) \cdot (G-2) \cdot MTTR^2} = \frac{2 \cdot 1515^3}{6 \cdot 5 \cdot 4 \cdot 15^2} = 257575, 25 = 705 years$

Question 20 If the same type of disk is used in a RAID0 architecture, what is the maximum number of disks to be used to have a MTTDL > 1.5 years?

- [A] None
- [B] 1
- [C] 3
- [D] 2

SOLUTION:
MTTDL is computed as:
$\frac{MTTF}{N} = \frac{1515}{2} = 2.07$

Performance - (8 points)

Let us consider a computing infrastructure that consists of Servers A,B and C and that can be accessed by a large number of users. In order to evaluate the network performance, a resources monitoring procedure was set-up for $T = 15$ minutes and the results are the following:

- The number of system completions is: $C = 1080$
- The throughput of the servers are: $X_A = 2.16 job/s$, $X_B = 1.68 job/s$, $X_C = 2.16 job/s$
- The service time of the servers are: $S_A = 100ms$, $S_B = 500ms$, $S_C = 450ms$

1. Define the system model assuming that each server is modelled by a single service center and neglecting the possible network components
2. Compute
   (a) the system throughput X.
   (b) the number of visits of each server
   (c) the service demand of each server
   (d) the utilization of each server. Which server must be improved to increase the system throughput. Motivate your answer.
3. In this system is it possible obtaining a maximum throughput of 8 job/sec? If not, which kind of changes would you suggest to the system architect to obtain it? (Please motivate your answer and prove the validity of your point with computations showing the characteristics of the (possible) new servers)
4. Let us now consider the original system with a fixed number N of users and Z=1 sec.
   (a) If we have N=2, the system throughput could be equal to $1j/s$? Motivate your answer.
   (b) Is it possible to observe a system response time R < 20 sec? At which conditions?

SOLUTION:

1) We can define an open model like the one in Figure

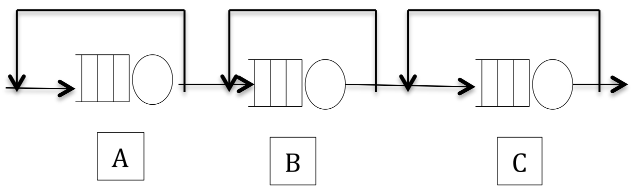


Figure 3: QN model

2)

a) $T = 15min = 900sec$, X= C/T, X=1080/900=1.2 j/s

b) Forced flow law: $X_k = V_k * X$, so $V_k = X_k/X$,
$V_A = 2.16/1.2 = 1.8$

$V_B = 1.68/1.2 = 1.4$
$V_C = 2.16/1.2 = 1.8$

c) $D_k = V_k \cdot S_k$:
$D_A = 1.8 * 100 = 180ms = 0.18s$,
$D_B = 1.4 * 500 = 700ms = 0.7s$,
$D_C = 1.8 * 450 = 810ms = 0.81s$

d) $U_k = X \cdot D_k$:
$U_A = 1.2 * 0.18 = 0.216$,
$U_B = 1.2 * 0.7 = 0.84$,
$U_C = 1.2 * 0.81 = 0.972$, Bottleneck server C

3)

The original system has the following characteristics: $X_{\max} = \frac{1}{D_{\max}} = 1.235job/s$, so it is not possible obtaining a throughput of 8 job/sec. To have $X_{\max} = 8j/s$ the system should have a $D_{\max} = 0.125$. This implies that the three servers A,B and C should be all substituted with new ones. For example: Server A with a new A' twice as faster, so with $D'_A = 0.09$ and Server B with a new B' six times faster, so with $D'_B = 0.1166$, Server C with a new C' seven as faster, so with $D'_C = 0.1157$.

4)
a) $N^* = \frac{D+Z}{D_{\max}} = \frac{2.69}{0.81} = 3.32$. Since $N < N^*$, the system is in light-load condition and $X_{\max} = \frac{N}{D+Z} = 0.74\ j/s$, so it is not possible observing a throughput of $1j/s$

b) It is a closed system, so the bounds for closed models can be applied. To guarantee the possibility that R could be <20 sec, we can work on the lower bound and guarantee that max(D, NDmax-Z)<R(N)<20. So N*0.81<19, the possibility is given for N< 19/0.81=23.46. Working on the upper bound ND<20 we would obtain N< 20/1.29=11.8.

Answer sheet:

Last Name / Cognome:

....

First Name / Nome:

....

Answers of the multiple-choice part of the exam must be given exclusively on this sheet

Student ID : ☐**0** ☐**1** ☐**2** ☐**3** ☐**4** ☐**5** ☐**6** ☐**7** ☐**8** ☐**9**
Student ID : ☐**0** ☐**1** ☐**2** ☐**3** ☐**4** ☐**5** ☐**6** ☐**7** ☐**8** ☐**9**
Student ID : ☐**0** ☐**1** ☐**2** ☐**3** ☐**4** ☐**5** ☐**6** ☐**7** ☐**8** ☐**9**
Student ID : ☐**0** ☐**1** ☐**2** ☐**3** ☐**4** ☐**5** ☐**6** ☐**7** ☐**8** ☐**9**
Student ID : ☐**0** ☐**1** ☐**2** ☐**3** ☐**4** ☐**5** ☐**6** ☐**7** ☐**8** ☐**9**
Student ID : ☐**0** ☐**1** ☐**2** ☐**3** ☐**4** ☐**5** ☐**6** ☐**7** ☐**8** ☐**9**

Big Data

Question 01 : ☐**A** ☐**B** ☐**C** ☐**D**
Question 02 : ☐**A** ☐**B** ☐**C** ☐**D**
Question 03 : ☐**A** ☐**B** ☐**C** ☐**D**
Question 04 : ☐**A** ☐**B** ☐**C** ☐**D**

Dependability

Question 05 : ☐**A** ☐**B** ☐**C** ☐**D**
Question 06 : ☐**A** ☐**B** ☐**C** ☐**D**
Question 07 : ☐**A** ☐**B** ☐**C** ☐**D**
Question 08 : ☐**A** ☐**B** ☐**C** ☐**D**
Question 09 : ☐**A** ☐**B** ☐**C** ☐**D**
Question 10 : ☐**A** ☐**B** ☐**C** ☐**D**

Cloud and Virtualization

Question 11 : ☐**A** ☐**B** ☐**C** ☐**D**
Question 12 : ☐**A** ☐**B** ☐**C** ☐**D**
Question 13 : ☐**A** ☐**B** ☐**C** ☐**D**
Question 14 : ☐**A** ☐**B** ☐**C** ☐**D**
Question 15 : ☐**A** ☐**B** ☐**C** ☐**D**
Question 16 : ☐**A** ☐**B** ☐**C** ☐**D**

Dsks

Question 17 : ☐**A** ☐**B** ☐**C** ☐**D**
Question 18 : ☐**A** ☐**B** ☐**C** ☐**D**
Question 19 : ☐**A** ☐**B** ☐**C** ☐**D**
Question 20 : ☐**A** ☐**B** ☐**C** ☐**D**